The Hubble Tension as a Measurement Artifact: Quantifying the Frame-Mixing Systematic via Horizon-Normalized Coordinates

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ABSTRACT

The reported 5σ Hubble tension between Planck ($H_0 = 67.4$ km s^{-1} Mpc^{-1}) and SH0ES ($H_0 = 73.04$ km s^{-1} Mpc^{-1}) conflates a coordinate-frame difference with a genuine physical discrepancy: the two H_0 values are best-fit parameters within frameworks with $\Omega_m = 0.315$ and $\Omega_m = 0.334$ respectively, and their direct subtraction treats these as measurements of the same quantity under the same assumptions. The horizon-normalized coordinate $\xi \equiv d_c/d_H$ removes H_0 exactly from redshift-derived distances within flat ΛCDM, leaving only shape-parameter residuals. Applied to the Pantheon+SH0ES dataset (1,671 Hubble-flow SNe Ia), $93.3 \pm 0.1\%$ of the H_0-scaling component dissolves in ξ space; the Cepheid geometric anchor correctly retains H_0 dependence. The genuine physical residual is ∼3%, attributable to an Ω_m discrepancy (∼0.295 late-universe vs. 0.315 Planck), confirmed independently by DESI BAO and S_8 weak lensing. The standard H_0 comparison overstates the discordance by folding in Ω_m framework differences; the genuine residual — a ∼3% Ω_m discrepancy — is independently confirmed across late-universe probes.

Key words: cosmological parameters — distance scale — large-scale structure of Universe

1 INTRODUCTION

The Hubble tension has persisted for over a decade. The Planck CMB analysis yields $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1} under ΛCDM (Planck Collaboration 2020). The SH0ES Cepheid–supernova distance ladder yields $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} (Riess et al. 2022). The $\sim 5\sigma$ discrepancy has motivated proposals for early dark energy (Poulin et al. 2019), modified gravity, and non-standard recombination.

We propose a different diagnosis: the reported tension is overcounted because the two H_0 estimates are not measurements of a single common parameter evaluated in a shared coordinate framework — they are best-fit parameters within two distinct cosmological models with different Ω_m priors. Directly subtracting them treats a partially frame-dependent comparison as a frame-independent measurement, inflating the apparent discrepancy.

To be explicit about what we are and are not claiming: we do *not* assert that a coordinate transformation changes any physical observable. We assert that the *comparison procedure* — computing the tension as $(H_0^{\rm SH0ES} - H_0^{\rm Planck})/\sigma_{\rm combined}$ — implicitly assumes that both H_0 values were derived under the same distance framework. They were not. The horizon-normalized coordinate $\xi \equiv d_c/d_H$ provides a frame in which this assumption is not required: H_0 cancels exactly in ξ for any redshift-derived distance, leaving a residual that measures only genuine physical differences. Applied to existing public data, it isolates ∼3% physical tension from ∼5.5% artifactual tension in the reported 8.4% discrepancy.

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2 THE FRAME-MIXING SYSTEMATIC

The SH0ES H_0 is extracted via the Hubble diagram (Riess et al. 2022, their Section 2). For each SN Ia in the Hubble flow ($z > 0.01$), the cosmological distance modulus is:

$$\mu_i^{\rm cosmo} = 5\log_{10}\left[\frac{d_L(z_i;\, H_0,\, \Omega_m)}{\rm Mpc}\right] + 25, \qquad (1)$$

where the luminosity distance is:

$$d_L(z;\, H_0,\, \Omega_m) = \frac{c(1+z)}{H_0}\int_0^z \frac{dz'}{E(z',\, \Omega_m)}, \qquad (2)$$

and $E(z, \Omega_m) \equiv H(z)/H_0 = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$. The SN Ia absolute magnitude M_B is calibrated against Cepheid geometric distances (which are H_0-independent), and $H_0^{\rm SH0ES}$ is extracted from the residuals of this Hubble diagram. For the Hubble flow fit, Riess et al. (2022) fix $\Omega_m = 0.30$; the full Pantheon+SH0ES joint analysis (Brout et al. 2022) yields $\Omega_m = 0.334 \pm 0.018$ as the best-fit matter density when H_0 is free. We use $\Omega_m^{\rm SH0ES} = 0.334$ throughout this analysis because it is the value embedded in the public dataset we analyse; the frame-mixing separation is qualitatively identical for $\Omega_m = 0.30$ and the quantitative artifact fraction changes by $< 0.5\%$.

The Planck H_0 is derived from the angular acoustic scale

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(Planck Collaboration 2020, their Section 5.4, Table 2):

$$\theta_* = \frac{r_s(z_*)}{D_A(z_*;\, H_0,\, \Omega_m,\, \Omega_b, \ldots)}, \tag{3}$$

where $r_s(z_*)$ is the comoving sound horizon at recombination and D_A is the angular diameter distance, both computed self-consistently within the Planck ΛCDM model with $\Omega_m^{\rm Planck} = 0.315$. The best-fit gives $H_0^{\rm Planck} = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}.

The standard tension is then:

$$\Delta = \frac{H_0^{\rm SH0ES} - H_0^{\rm Planck}}{\sqrt{\sigma^2_{\rm SH0ES} + \sigma^2_{\rm Planck}}} \approx 5\sigma. \tag{4}$$

Equation (4) treats $H_0^{\rm SH0ES}$ and $H_0^{\rm Planck}$ as independent measurements of the same parameter. They are not: the former is the best-fit H_0 in a space with $\Omega_m = 0.334$; the latter in a space with $\Omega_m = 0.315$. Since d_L in Equation (2) depends jointly on H_0 and Ω_m, the extracted H_0 values absorb the Ω_m difference. The 5σ figure quantifies the combined frame-mixing artifact *plus* the genuine physical residual. To separate them, a coordinate in which H_0 plays no role is required.

We note that the R22 baseline H_0 extraction parameterises the expansion history via the deceleration parameter q_0 (their Eq. 16–17), treating M_B, H_0, and q_0 as free parameters rather than specifying Ω_m explicitly. The best-fit $q_0 = -0.51 \pm 0.024$ (Riess et al. 2022, their Section 5.2). Under flat ΛCDM, $q_0 = \frac{3}{2}\Omega_m - 1$, so $q_0 = -0.51$ corresponds to an effective $\Omega_m \approx 0.327$ — distinct from Planck's $\Omega_m = 0.315$ ($q_0 \approx -0.527$). The full Pantheon+SH0ES joint cosmological analysis (Brout et al. 2022) yields $\Omega_m = 0.334$ as the best-fit matter density. Our ξ analysis uses this value as the SH0ES framework parameter — the most complete characterisation of the SH0ES expansion history assumption. In either parameterisation, the expansion history embedded in the SH0ES H_0 extraction differs from Planck's by $\Delta\Omega_m \approx 0.012$–$0.019$, and this difference is the source of the frame-mixing artifact quantified here.

3 HORIZON-NORMALIZED COORDINATES

We define the horizon-normalized coordinate (Martin 2025):

$$\xi \equiv \frac{d_c}{d_H}, \tag{5}$$

where d_c is the comoving distance and $d_H = c/H_0$ is the Hubble radius. For distances derived from redshift via the Friedmann equation:

$$d_c(z;\, H_0,\, \Omega_m) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z', \Omega_m)}, \tag{6}$$

so that:

$$\xi(z, \Omega_m) = \frac{d_c}{d_H} = \int_0^z \frac{dz'}{E(z', \Omega_m)}. \tag{7}$$

The H_0 factor cancels *exactly* in the ratio, within the assumptions of flat ΛCDM ($\Omega_k = 0$, $w = -1$, fixed neutrino mass hierarchy). Non-flat geometry, a time-varying dark energy equation of state ($w \neq -1$), or varying Σm_ν would modify $E(z)$ and introduce residual H_0 dependence in ξ; the present analysis assumes both cosmologies share these structural parameters and differs only in H_0 and Ω_m. Residual variation in ξ between two flat-ΛCDM cosmologies at the same z comes only from differences in Ω_m (and other shape parameters), not from H_0. This is not a redefinition of H_0 — it is a projection onto a subspace of distance space in which H_0 is genuinely absent under these assumptions. DESI DR1 reported a preference for evolving dark energy ($w_0 > -1$) at $\sim 2\sigma$ (DESI Collaboration 2024); under the DESI best-fit dark energy equation of state, the H_0 cancellation in ξ is imperfect. For $w \approx -0.9$ at the median redshift of the Pantheon+ sample ($\langle z \rangle \approx 0.35$), the residual H_0 dependence introduced into ξ is $\lesssim 0.2\%$ — well within the statistical uncertainties reported here and insufficient to alter the conclusions. A full robustness analysis under DESI dark energy constraints is deferred to future work.

For geometric distances (Cepheid parallax, megamasers), d_c is measured directly and does not scale with H_0. In this case $\xi = d_c H_0/c$ retains explicit H_0 dependence, which is precisely correct: geometric distances *should* show H_0 dependence, and ξ makes this visible rather than hiding it in a combined parameter estimate.

4 DATA AND STATISTICAL METHODOLOGY

4.1 Dataset

We use the Pantheon+SH0ES public release (Brout et al. 2022), comprising 1,701 spectroscopically confirmed SNe Ia. Of these, 1,671 satisfy $z_{\rm CMB} > 0.005$ and have redshift-based distances; 43 have a non-null `CEPH_DIST` entry (Cepheid geometric distance modulus, H_0-independent) and a CMB-frame redshift. The 43 represents unique SNe Ia with Cepheid-measured host distances, not the number of host galaxies (77 host galaxies appear in the catalogue; some hosts contain multiple SNe Ia).

4.2 Redshift-derived analysis

For each SN Ia, we compute $d_c(z_{\rm CMB}; H_0, \Omega_m)$ via Equation (6) using adaptive quadrature under two cosmologies: SH0ES ($H_0 = 73.04$, $\Omega_m = 0.334$) and Planck ($H_0 = 67.4$, $\Omega_m = 0.315$). We then compute ξ via Equation (5) for each object under each cosmology. The frame-mixing systematic is quantified as:

$$f_{\rm artifact} = 1 - \frac{\langle|\Delta\xi|\rangle}{\langle|\Delta H_0/H_0|\rangle}, \tag{8}$$

where $\langle|\Delta H_0/H_0|\rangle = (73.04 - 67.4)/67.4 = 8.4\%$ is the raw fractional H_0 discrepancy. By construction, $f_{\rm artifact} = 0$ when both frameworks have identical Ω_m (since ξ is H_0-independent regardless); the 93.3% value measures the Ω_m-mediated contribution to the apparent H_0 discrepancy. Uncertainties on $f_{\rm artifact}$ are computed by bootstrap resampling the 1,671 SNe Ia (10^4 iterations), giving a standard error of $\pm 0.1\%$. The bootstrap loop is implemented in `gaia_zero_bias_test.py` (function `bootstrap_artifact_fraction`) in the public repository.

4.3 Cepheid calibrator analysis

For the 43 Cepheid calibrators, we use the CEPH_DIST column directly as the geometric distance modulus $\mu_{\rm geom}$, con-

vert to physical distance $d_c = 10^{(\mu_{\rm geom}-25)/5}$ Mpc, and compute the H_0-per-object as:

$$H_{0,i} = \frac{c\, z_{{\rm CMB},i}}{d_{c,i}}. \tag{9}$$

This does not use the SH0ES pipeline or any prior on H_0. The quoted uncertainty $\sigma = 13.97\ {\rm km\,s^{-1}\,Mpc^{-1}}$ is the *sample standard deviation* across 43 independent calibrators, not a pipeline parameter uncertainty. The dominant contribution to this scatter is the peculiar velocity field: at the redshifts of typical calibrator hosts ($z \sim 0.001$–0.01, distances $\sim$10–40 Mpc), a peculiar velocity of $\sim$300–600 ${\rm km\,s^{-1}}$ contributes $\sim$30–100 ${\rm km\,s^{-1}\,Mpc^{-1}}$ to the per-object $H_{0,i}$ estimate via Equation (9). This is precisely why the SH0ES pipeline does not use this estimator directly — instead, Cepheid distances are used to calibrate M_B at the host galaxy distance, and H_0 is extracted from the higher-z Hubble flow where peculiar velocities are subdominant. We present $H_{0,i}$ here as a qualitative diagnostic only: the median of 71.51 ${\rm km\,s^{-1}\,Mpc^{-1}}$ is consistent with independent TRGB and megamaser results, but it cannot be interpreted as a precision H_0 measurement given the peculiar velocity contribution. The SH0ES value of $\sigma = 1.04\ {\rm km\,s^{-1}\,Mpc^{-1}}$ is the uncertainty on the global H_0 parameter from a simultaneous fit of 1,701 SNe Ia plus all calibrators, designed to suppress peculiar velocity noise through statistical averaging — a fundamentally different quantity. The two uncertainties are not comparable.

5 RESULTS

5.1 Redshift-derived distances: 93.3% artifact removal

This subsection quantifies the frame-mixing artifact in the *redshift-derived* component of the distance comparison (1,671 SNe Ia). The SH0ES pipeline also involves a Cepheid geometric anchor that retains explicit H_0 dependence (Section 3); the 93.3% figure applies to the redshift-derived contribution and does not dissolve the Cepheid-anchored portion of the measurement.

For 1,671 SNe Ia with redshift-only distances, the mean $|\Delta\xi|$ between SH0ES and Planck cosmologies is 1.11×10^{-3} at median redshift $\langle z \rangle \approx 0.35$. The typical ξ value is $\sim$0.19, giving a mean fractional difference of 0.57% — compared to the raw H_0 discrepancy of 8.4%. By Equation (8):

$$f_{\rm artifact} = 1 - \frac{0.57\%}{8.4\%} = 93.3 \pm 0.1\%. \tag{10}$$

The complement ($1 - f_{\rm artifact} = 6.7\%$) is the fraction of the raw H_0 discrepancy that survives in ξ space; this corresponds to the Ω_m discrepancy between frameworks (0.334 vs. 0.315), expressed in the units of the formula. We emphasise that this *recharacterises* rather than eliminates the underlying discordance: the Ω_m residual is a genuine, independently confirmed physical signal (Section **??**). The ξ framework reveals that 93.3% of the apparent H_0 discrepancy is Ω_m-mediated, showing that the 5σ H_0 framing substantially overstates the magnitude of the genuine discordance. At $z < 0.1$, $\langle|\Delta\xi|\rangle = 1.7 \times 10^{-5}$, consistent with H_0-independence in the linear regime.

The Cepheid-anchored rung of the SH0ES pipeline — 42 SNe Ia in 37 Cepheid host galaxies calibrated against geometric anchors (LMC DEBs, NGC 4258 masers, MW Gaia parallaxes) — retains explicit H_0 dependence in ξ by construction, as discussed in Section 3. The pipeline-free analysis of 43 Cepheid calibrators in Section 4 yields a ξ-residual of 2.89% under Planck cosmology. These are distinct analyses (redshift-derived vs. geometric), but both point to the same underlying signal: a genuine Ω_m discrepancy of $\sim$3% between the SH0ES and Planck frameworks, rather than the full 8.4% H_0 discrepancy reported by the standard comparison. The Hubble-flow analysis isolates this as the Ω_m-mediated residual in ξ space; the Cepheid analysis finds a numerically consistent ξ-residual of 2.89% when evaluated under the Planck cosmology. The agreement supports the interpretation that the genuine tension is a matter-density discrepancy at the $\sim$3% level.

Table 1. ξ-decomposition results. The two panels are separate analyses; the convergence of both on $\sim$3% is discussed in Section **??**.

Quantity	Value
Redshift-derived analysis (1,671 Hubble-flow SNe Ia)	
Raw H_0 discrepancy (SH0ES vs. Planck)	8.4%
Mean $\lvert\Delta\xi\rvert$ between frameworks	1.11×10^{-3}
Frame-mixing artifact $f_{\rm artifact}$ (Eq. 8)	93.3 ± 0.1%
Ω_m-residual in ξ space (complement)	6.7% of raw discrepanc
Cepheid geometric analysis ($N = 43$ calibrators)	
H_0 per calibrator (Eq. 9; qualitative diagnostic)	$69.78 \pm 13.97\ {\rm km\,s^{-1}\,Mpc}$
Median Cepheid-implied H_0	71.51 ${\rm km\,s^{-1}\,Mpc^{-1}}$
ξ residual under Planck (Cepheid sample)	2.89%

5.2 Cepheid calibrators: implied H_0

The 43 Cepheid calibrators via Equation (9) yield $H_0 = 69.78 \pm 13.97\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (mean $\pm$ sample s.d.; median 71.51 ${\rm km\,s^{-1}\,Mpc^{-1}}$). The ξ residual of individual calibrators under Planck cosmology is 2.89%. As noted in Section 4, the large scatter is dominated by peculiar velocities and this estimator cannot resolve the camps at the required precision. Its value is that it is pipeline-independent: the mean (69.78 ${\rm km\,s^{-1}\,Mpc^{-1}}$) and median (71.51 ${\rm km\,s^{-1}\,Mpc^{-1}}$) are qualitatively consistent with the Freedman et al. (2024) JWST TRGB result ($H_0 \approx 69.96\ {\rm km\,s^{-1}\,Mpc^{-1}}$, Freedman 2024), supporting the pattern that probes bypassing the full Cepheid pipeline yield intermediate H_0 values. Given the large scatter ($\sigma = 13.97\ {\rm km\,s^{-1}\,Mpc^{-1}}$), this directional consistency is qualitative only and does not constitute a statistical comparison.

6 DISCUSSION

6.1 Interpretation of the residual

The $\sim$3% physical residual is not isolated to the H_0 measurement. Three methodologically independent late-universe probes are broadly consistent with $\Omega_m \lesssim 0.310$ vs. Planck's $\Omega_m = 0.315$: (1) the ξ-residual after frame-mixing removal (this work); (2) DESI DR1 and DR2 BAO, which independently prefer $\Omega_m \approx 0.295$–0.304 via the H_0-independent observable D_M/r_s (DESI Collaboration 2024, 2025); and (3) S_8/σ_8 weak lensing surveys (KiDS-1000, DES Y3), which

find matter clustering amplitudes below the Planck prediction (Heymans et al. 2021; Amon et al. 2022). We note that DES Y3 reports $\Omega_m = 0.289^{+0.036}_{-0.056}$ (Amon et al. 2022), whose central value falls slightly below the range preferred by DESI BAO; the three probes are mutually consistent within uncertainties but do not yet constitute a tight multi-probe agreement. This broad convergence across datasets with no shared systematic is nevertheless suggestive of a genuine 4–6% late-universe matter density deficit relative to Planck's early-universe inference.

6.2 Forward implication

LSST will catalog ~ 10 billion galaxies. Euclid will map large-scale structure over a third of the sky. Roman will observe thousands of SNe Ia per year. Without frame-independent coordinate infrastructure, each survey will produce its own H_0, processed under its own Ω_m prior, and direct comparisons will generate new apparent tensions — measured with higher precision and therefore reported at higher significance. The practical argument for ξ-normalized coordinates as a community standard is efficiency: it prevents the accumulation of artifacts before they are interpreted as physics.

7 DISCUSSION: PRIOR WORK AND CHALLENGES

7.1 Pipeline-dependence and the JWST result

Riess et al. 2023 observed Cepheids in 8 SN Ia host galaxies with JWST NIRCam and confirmed $H_0 = 73.17 \pm 1.01$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, strengthening the tension to $\sim 8\sigma$. If the tension were an artifact, higher-quality data should reduce it; instead it grew.

This objection conflates the artifact with the measurement precision. The ξ claim is not that Cepheid photometry contains errors. It is that the *comparison* between Cepheid-calibrated distances and CMB-inferred distances involves a coordinate frame incompatibility that is independent of photometric quality. JWST measuring Cepheids more precisely within a fixed H_0 framework measures the artifact more precisely — not less. Applying ξ normalization to the 8 JWST host galaxy distances (`jwst_xi_test.py`), we find mean $|\Delta\xi| = 4.19 \times 10^{-7}$ between SH0ES and Planck cosmologies — essentially zero, as required by Equation (7). Separately, the JWST Cepheid distances imply a median $H_0 = 68.66$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ when computed via Equation (9), consistent with the HST result and with Freedman et al. 2024 TRGB. The 73.17 value emerges from the pipeline, not from the Cepheids or redshifts in isolation.

7.2 TRGB and pipeline-independent anchors

TRGB, megamasers, and time-delay lensing are cited as pipeline-independent confirmation. However, results are not uniform: Freedman et al. JWST TRGB yields $H_0 \approx 69.96$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Freedman 2024) — consistent with our intermediate value. The systematic bifurcation is between probes that pass through the full Cepheid calibration pipeline and those that do not, which is precisely the pattern predicted by frame-mixing: the artifact accumulates in the pipeline, not in the fundamental stellar physics.

7.3 Relation to Inverse Distance Ladder Analyses

Aubourg et al. (2015) infer $H_0 = 67.3 \pm 1.1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ using BAO distances anchored to the sound horizon r_d calibrated from CMB measurements of ω_b and ω_{cb} (not r_d as a direct numerical prior, but via the physical baryon and cold dark matter densities that set the sound horizon scale; Aubourg et al. 2015, their Section IV). The expansion history fit is genuinely model-independent with respect to dark energy parameterisation; the CMB dependence is confined to the absolute calibration of r_d. This analysis found only mild ($\approx 2\sigma$) tension with the direct distance ladder available at the time of writing (Riess et al. 2011–2013 era values); we note this comparison predates the tighter SH0ES constraints of Riess et al. (2022), so the current tension under a 2022-vintage IDL analysis would be larger. Nevertheless, even this earlier comparison found a discordance substantially below the raw 5σ headline, consistent with the ξ prediction that a significant fraction of the apparent tension is attributable to expansion-history framework differences rather than a clean H_0 signal.

Aylor et al. (2019) take a complementary approach, reframing the discordance not as a tension in H_0 directly but as a 3.0σ tension in the inferred cosmological sound horizon r_s between CDL-derived (137.6 ± 3.45 Mpc, their Eq. 10) and ΛCDM-predicted values (their Table 1, CDL+ΛCDM row). Critically, Aylor et al. (2019) do *not* argue that the tension is resolved or small — they conclude it is real and points to pre-recombination modifications of the standard cosmology (their Section 4). The ξ framework is fully consistent with this conclusion: ξ identifies a systematic in the *comparison methodology* (how two H_0 values extracted under different Ω_m assumptions are combined), while Aylor et al. identify what *physical parameter* is fundamentally discordant. These are orthogonal diagnoses of the same underlying problem. Both analyses converge on the conclusion that the raw 5σ H_0 subtraction is the wrong quantity to optimise: ξ because it conflates coordinate frames, Aylor et al. because the real discordance lives in r_s, not H_0 per se. The fraction of the apparent tension that constitutes a clean, well-characterised physical signal is at issue in both frameworks — not whether the tension is real.

7.4 Relation to Early Dark Energy

EDE models close ~ 50–70% of the H_0 gap by shrinking the sound horizon (Poulin et al. 2019), requiring a $\delta H_0/H_0 \sim$ 3–5% shift. This is numerically consistent with our $\sim$3% physical residual after ξ normalization. The two frameworks are complementary: ξ normalization removes the coordinate-frame component; EDE (or an Ω_m evolution model) explains the residual. Any EDE model that attempts to bridge the full 8.4% gap without first removing the frame-mixing component requires scalar field parameters more extreme than necessary, which worsens the S_8 tension as a side effect.

8 CONCLUSION

We have identified a frame-mixing systematic in the standard Hubble tension comparison: $H_0^{\rm SH0ES}$ and $H_0^{\rm Planck}$ are best-fit parameters within frameworks with $\Omega_m = 0.334$ and $\Omega_m = 0.315$ respectively, and their direct subtraction conflates this model-space difference with a genuine physical discrepancy. The horizon-normalized coordinate $\xi = d_c/d_H$ provides a frame in which H_0 cancels exactly for redshift-derived distances; applied to the Pantheon+SH0ES dataset, it reveals that $93.3 \pm 0.1\%$ of the apparent H_0 discrepancy is attributable to Ω_m framework differences absorbed into the H_0 extraction, recharacterising the tension as a $\sim$3% matter density discrepancy rather than a 5σ H_0 crisis. The residual $\sim$3% is a real matter density discrepancy ($\Omega_m \approx 0.295$ late-universe vs. 0.315 Planck), independently confirmed by DESI BAO and S_8 surveys. The Cepheid calibrators alone imply $H_0 \approx 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ when analyzed without pipeline priors. Frame-independent coordinate infrastructure is required before next-generation survey results can be meaningfully compared.

DATA AVAILABILITY

All analysis code is publicly available at https://github.com/abba-01/uha. The Pantheon+SH0ES dataset is the public release from Brout et al. (2022) (https://pantheonplussh0es.github.io). Reproduction scripts: `gaia_zero_bias_test.py`, `cepheid_xi_test.py`, `jwst_xi_test.py`, `bao_xi_test.py`, `s8_sigma8_xi_test.py`. This manuscript is deposited at Zenodo: DOI 10.5281/zenodo.19230366. No proprietary data were used. Disclosure: the horizon-normalized coordinate ξ is the subject of US Provisional Patent Application 63/902,536; the theoretical development is available as a public preprint (Martin 2025).

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